ARAB ACADEMY FOR SCIENCE, TECHNOLOGY AND MARITIME TRANSPORT

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AI-Driven Predictive Maintenance and Asset Reliability Decision-Support System

Comprehensive Analytical Report · Capstone Project · Team 3

Team members	Ahmed Fawzy Mohamed Rashed (24220998) Ahmed Abd Elhakam Abd Elfattaah Zaian (24224008) Khaled Abdelnasser Morsi Amer (24221044) Malak Philippe Sadek Reiad (24220177) Ali Elraey (24126750)
Instructors	Dr. Yasser Yakout Dr. Hossam Daoud
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Companion files: Group_9226_Team_3_Dashboard.pbix and Group_9226_Team_3_Presentation.pptx.
All results are based on a synthetic educational dataset and do not describe a real company.

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1. Executive Summary

Our team was asked to act as an analytics advisory group for a company that maintains a large fleet of equipment: heavy machinery, railway signaling, medical equipment, industrial pumps and power systems. The company records the maintenance work it completes, but it cannot easily see which assets are likely to fail next, how much a failure really costs, or which jobs should come first. Our executive decision domain covers asset condition, failures, work orders, downtime and economic impact. The decision we support is therefore: **where should maintenance, technician and spare-parts effort be focused to reduce downtime and lost production?**

We worked with the assigned dataset of 5,000 records covering January 2024 to December 2026. After checking the data quality, we explored the data, built a Power BI dashboard and tested whether a machine learning model could help rank risk. Our main findings are:

1. **Downtime costs far more than maintenance.** Over three years the company spent $50.4 million on maintenance but lost $228.9 million of production because equipment was down. That is about $4.50 of lost production for every dollar spent on repairs, and roughly $2,513 for every hour of downtime.
2. **The problem is stable, not getting worse.** Downtime and losses stay at about the same level in every quarter, so today's performance is the baseline any improvement must beat.
3. **Critical assets take too long to repair.** They do not break down more often than other assets, but each repair takes 16.3 hours on average against about 11 hours for the rest, and only 29% of their jobs meet the agreed service time.
4. **Reactive maintenance (fixing things only after they break) is the most expensive approach.** These records lose about $62,000 each compared with about $40,000 under condition-based or predictive maintenance. The gap gets smaller but does not disappear when we take asset age and condition into account.
5. **Older and worn assets cause most of the downtime.** Assets older than 12 years lose about three times more than assets younger than four years.
6. **A simple machine learning model can help planners prioritize.** Our model predicts which jobs are likely to become high-loss events better than a simple rule based on asset type and criticality (a score of 0.73 against 0.61, where 0.5 is a coin flip). The 28% of records it flags carry 44% of all production loss. However, it also raises many false alarms, so it should support people, not replace them.

Our recommendations are to speed up repairs on critical assets, fix a few data definitions, and set financial reserves based on large events, all of which can start immediately. We also recommend a six-month controlled trial of condition-based maintenance and of the model's watchlist before any larger investment.

Financial benefits. Our simulation (Section 10.2) estimates that, in the expected scenario, these actions could avoid about 2,766 hours of downtime a year, giving a net benefit of about $6.56M a year after costs and a three-year net present value of about $16.31M (range $7.22M to $28.06M across our scenarios). These figures are estimates, not achieved savings.

Limits. The data is synthetic, so our results show patterns inside a simulation, not proven cause and effect. Any savings we mention are estimates that must be confirmed by a real trial.

2. Introduction and Business Context

2.1 The business setting

The company in our case runs five types of assets, each in a different industry: heavy equipment in mining, industrial pumps in oil and gas, railway signaling, biomedical equipment in healthcare, and power systems in infrastructure. Assets are maintained using four strategies, from purely reactive (repair after failure) to predictive (act on signs of coming failure). Over the three years in our data the fleet recorded 7,597 failures and more than 91,000 hours of downtime.

For organizations like this, an equipment failure is not only a repair bill. It stops production, delays trains or interrupts hospital services. The real question for managers is therefore not "how much do we spend on maintenance?" but "are we spending it in the places that prevent the most expensive downtime?"

2.2 Why analytics: from describing to prescribing

In the course we studied the analytics maturity model, in which organizations move from *descriptive* analytics (what happened), to *diagnostic* (why it happened), *predictive* (what will happen) and finally *prescriptive* analytics (what we should do). Our project follows the same path. The dashboard and the exploratory analysis are descriptive and diagnostic. The machine learning model is predictive. Our recommendations are a first, careful step towards prescriptive decisions, but we do not claim to have found an "optimal" plan.

2.3 Stakeholders

Stakeholder	What they decide	What they need from us
Maintenance and Reliability Director	Maintenance strategy and where to invest	A fair comparison of strategies and the main causes of downtime
Maintenance planners	Which jobs come first each day	A ranked list of high-risk work
Operations managers	Acceptable downtime and repair targets	Repair times and service performance by asset
Finance	Maintenance budget and reserves	The cost of downtime compared with spending
Procurement	Spare-parts stock for critical assets	Where parts shortages hurt most

2.4 Scope

We cover all 5,000 records, the dashboard, three machine learning methods that we actually ran, and our recommendations. We do not analyze individual assets or sites, because the asset and site codes in the data turned out not to identify the same physical asset over time (see Section 5).

2.5 Our analytical approach

Group 9226 is made up of five project teams, each with its own scenario: enterprise financial performance, quality in asset-intensive projects, predictive maintenance, resource allocation and digital transformation. As Team 3, we apply the same analytical principles as the other teams, but we keep the operating and financial definitions of our own predictive maintenance scenario, for example how downtime, failures and production loss are measured.

Our assigned project theme is the *AI-Driven Predictive Maintenance and Asset Reliability Decision-Support System*, and our executive decision domain covers five areas: **condition, failures, work orders, downtime and economic impact**. Table 2.2 shows where each area is analyzed.

Table 2.2 Our executive decision domain and where it is covered

Domain area	Main measures	Report sections	Dashboard page
Condition	Condition score, vibration, asset age, condition profiles	6.7, 6.8, 8.3	Reliability, Risk
Failures	Failures, MTBF, failure modes, high-loss risk	6.4–6.6, 8.1	Reliability, Risk
Work orders	Overdue rate, first-time fix, repeat repair, service target met	5.3, 6.5, 6.8	Work Orders
Downtime	Downtime hours, MTTR, availability, downtime per 1,000 planned hours	6.2, 6.5, 6.7, 8.2, 8.4	Summary, Reliability
Economic impact	Maintenance cost, production loss, cost per downtime hour	6.3, 6.6, 10.2	Cost & Loss

We followed the decision-focused approach set out in the capstone description, which asks teams to "begin with the decisions and business questions, establish trustworthy measures, examine performance and evaluate suitable AI applications" (Shehata & Daoud, 2026). The report is organized in the same order:

Step in the approach	What we did	Section
1. Start with the decisions and business questions	Defined the management decision, the stakeholders and eight business questions before touching the data	2, 3
2. Establish trustworthy measures	Checked data quality, documented problems, and defined every KPI from totals	4, 5
3. Examine performance	Explored trends, costs, asset classes, criticality, strategy, age and condition; built the dashboard	6, 7
4. Evaluate suitable AI applications	Tested four methods against simple baselines and reported where they did and did not add value	8
5. Turn evidence into decisions	Linked each insight to a recommendation with an owner, constraints and a next step	9, 10

We also agree with the guidance that analytical complexity is only useful when it helps answer the business questions and when the team can explain its results and limitations. For this reason we chose a simple, explainable model over a more complex one that did not perform better, and we report results where a model added little value (Sections 8.2 and 8.4) as openly as results where it helped. Appendix E shows where each learning outcome of the capstone is covered in this report.

2.6 How our findings should be read

The dataset we were given is a fully synthetic educational panel. It contains built-in business rules, operational relationships and derived indicators that were designed to support analytical exercises; they are not observed outcomes of a real company. We therefore keep three principles in mind throughout this report:

- **Correlation is not causation.** When two variables move together, for example reactive maintenance and higher losses, this is an association within the simulated data, not proof that one causes the other.
- **Prediction is not savings.** A model that predicts risk well does not by itself show that money will be saved or that operations will improve. Benefits can only be confirmed through a real trial.

- **Our own work, not the reference.** The project publication supplied with the case explains the scenario and the dataset. We used it to understand the variables, but the validation, analysis, dashboard, model evaluation and recommendations in this report are our own. Where our figures differ from the publication (for example MTBF and service performance, see Section 5.4), we explain why.

3. Business Questions

Before starting the analysis we agreed on the questions management would need answered. Each question is linked to a decision and to the part of our work that answers it.

No.	Business question	Decision it supports	Answered in
Q1	How large is the cost of downtime compared with maintenance spending?	How to set the budget	6.3; dashboard Cost & Loss
Q2	Is reliability getting better, worse or staying the same?	How urgent action is	6.2; 8.4
Q3	Which asset types and criticality levels account for most of the loss?	Where to focus	6.4, 6.5
Q4	Which maintenance strategy gives better results when we compare like with like?	Strategy mix	6.6
Q5	Which conditions (age, condition score, sensor readings) warn of failure?	When to inspect or overhaul	6.7, 6.8
Q6	Where is work execution weakest (overdue work, repeat repairs, service targets)?	Planning and staffing	6.5; dashboard Work Orders
Q7	Can a model rank jobs by risk better than a simple rule?	Whether to use a risk watchlist	8.1
Q8	Are there groups of assets with similar condition that need different care?	Targeted maintenance policies	8.3

Three practical challenges shaped how we answered these questions. First, the assets are very different, so we always compare rates (per hour, per failure, per record) rather than totals. Second, many variables are only known after a job is finished, such as downtime and cost, so a predictive model must not use them. Third, 2026 contains only half as many records as the earlier years, so years can only be compared using rates.

4. Dataset Description

4.1 Overview

Item	Description
Source	Group 9226 Team 3 Enterprise Dataset, version 1.0, with the supporting analytics workbook and data dictionary

Item	Description
Size	5,000 records and 60 variables (18 descriptive, 30 numeric measures, 12 derived indicators)
Period	5 January 2024 to 25 December 2026 (2,004 records in 2024, 2,000 in 2025 and 996 in 2026)
One record is	One observation of an asset for a period, with its maintenance work, failures, costs and condition readings
Record types	4,015 records with corrective work (all failures and losses) and 985 records with preventive work only
Nature	Fully synthetic data created for teaching, with built-in business rules. It contains no real company records.

4.2 Main dimensions

Dimension	Values
Asset class	Heavy Equipment, Industrial Pump, Railway Signaling, Biomedical Equipment, Power System (each linked to one industry)
Criticality	Critical (862 records), High (1,535), Medium (1,838), Low (765)
Maintenance strategy	Reactive (798), Preventive (1,754), Condition Based (1,345), Predictive (1,103)
Failure mode	Hydraulic leak, sensor failure, software/control, mechanical wear, electrical fault, or no failure
Other	Region (4), site (25), manufacturer (4), technician team (4), shift (day/night), condition status, risk category

4.3 Groups of variables and when they are known

Because we want to use the data for prediction, we sorted the variables by *when* they become known. This was important for avoiding "data leakage", which is when a model is accidentally given information about the result it is trying to predict.

Group	Examples	Known
Asset and planning information	Asset class, criticality, strategy, age, planned hours, service target, parts lead time	Before the work
Condition readings	Condition score, vibration, temperature, pressure, energy use	Before the work
Reliability results	Failures, downtime, availability, MTBF, MTTR	After the work
Work execution	Work-order counts, overdue work, first-time fixes, repeat repairs, actual service hours	After the work
Cost and consequences	Maintenance, parts and labor cost, production loss, stockouts, safety incidents	After the work
Derived indicators	Asset Health Index, Failure Probability, Maintenance Priority Score and nine others	Mixed; some include results

Two reliability terms appear throughout this report. **MTBF** (mean time between failures) is the average number of operating hours between breakdowns; higher is better. **MTTR** (mean time to repair) is the

average number of hours an asset is down for each failure; lower is better. Other terms are explained in Appendix B.

5. Methodology and Governance

5.1 Our approach

We kept the original CSV file unchanged and did all our work on a copy, using Python for the analysis and Power Query for the dashboard. We followed six steps: (1) check the structure of the data, (2) check the business rules, (3) define the KPIs, (4) explore the data, (5) build and test the models, and (6) agree recommendations and check that all three deliverables use the same numbers. This follows the data governance and quality practices discussed in the course: before analysis can be trusted, the data must be complete, consistent and well defined.

5.2 Data quality checks

Table 5.1 Summary of checks

Check	What we tested	Result
Completeness	Missing values in any of the 300,000 cells	None found
Uniqueness	Duplicate record IDs or duplicate rows	None found
Dates	Quarter field agrees with the record date	All agree
Availability and utilization	Recalculated from downtime, operating and planned hours	All agree
MTBF	Operating hours divided by failures	All agree
Work orders	Completed plus overdue equals total work orders	All agree
Repairs	First-time fixes plus repeat repairs equals corrective work orders	All agree
Costs	Maintenance cost is at least parts plus labor	All agree
Ranges	No negative values; percentages between 0 and 100%	All agree
Totals	Our totals compared with the workbook's KPI sheet	Exact match

5.3 Problems we found and how we handled them

- **Asset codes are not real assets.** Every asset code appears under all five asset classes, and every site appears in several regions. We therefore did not count unique assets or analyze individual assets or sites.
- **The stored MTTR column is inconsistent.** In 4,000 records it does not equal downtime divided by failures (differences of up to 4 hours). We recalculated MTTR from its parts instead of using the column.
- **Two different service measures.** The supplied "SLA Compliance Score" averages 75.8%, but only 46.1% of records actually met their service-time target. We use the second figure because it answers the question managers ask: did we meet the target?

- **Unexplained cost.** $9.2 million (18%) of maintenance cost is neither parts nor labor. We show it as "other" instead of guessing where it belongs.
- **Overdue counts.** 3,950 records have at least one overdue work order, but the total number of overdue work orders is 5,940. We are careful to say which one we mean.
- **Partial year and rare events.** 2026 has half the records, and there are only 3 safety incidents and 84 parts stockouts, which is too few to model.

We did not change any source values. These points are recorded so that the instructors and the company can review them.

5.4 Why we calculate rates from totals

A simple average of row-level rates gives misleading results, because a record with few operating hours counts as much as a record with many. Table 5.2 shows the difference.

Table 5.2 Row averages compared with rates calculated from totals

Measure	Average of rows (workbook)	Calculated from totals (our method)
MTBF	342.5 hours	313.3 hours
MTTR	11.7 hours	12.0 hours
Availability	96.8%	97.0%
Service target met	75.8% (score)	46.1% (records meeting target)

Using row averages would make the fleet look about 9% more reliable than it really is. All the totals (maintenance cost, production loss, failures and downtime) match the workbook exactly.

5.5 New fields we created

For the analysis and the dashboard we added: year and year-quarter; a flag for whether the service target was met; "other" maintenance cost; age bands (under 4, 4–8, 8–12 and over 12 years); condition-score bands; a high-loss flag used by the model (the top 20% of production loss); and the model's risk score, risk tier and condition profile.

5.6 The reference publication and our own results

The project publication supplied with the case is a resource for understanding the scenario. We used its variable lists and its five suggested dashboard designs as a starting point, and we summarize them only where needed. Everything else in this report, including the data checks, the exploratory analysis, the dashboard as built, the model results and the recommendations, is our own completed work. When we re-checked the publication's headline figures against the data, several of them changed:

Table 5.3 Reference publication figures compared with our validated figures

Measure	Reference publication	Our validated figure	Reason for the difference
MTBF	342.5 hours	313.3 hours	Publication averages row values; we divide total hours by total failures
MTTR	11.7 hours	12.0 hours	Stored MTTR column is inconsistent; we recalculate from downtime and failures
Availability	96.8%	97.0%	Row average compared with calculation from totals

Measure	Reference publication	Our validated figure	Reason for the difference
SLA performance	75.8% compliance	46.1% of jobs met target	The supplied score gives partial credit; we count jobs that met the target
Overdue work	3,950 overdue work orders	3,950 records with overdue work; 5,940 overdue work orders	The publication's figure counts records, not work orders
Number of assets	180 assets	Not used	Asset codes do not identify the same physical asset (Section 5.3)
Dashboards and AI	Five suggested designs; AI applications listed as possibilities	Eight implemented pages; four methods tested against baselines	Sections 7 and 8 report what we actually built and measured

6. Exploratory Data Analysis

In this section we explore the data to understand what drives downtime and loss. Unless stated otherwise, results cover all 5,000 records from 2024 to 2026.

6.1 Distributions and outliers

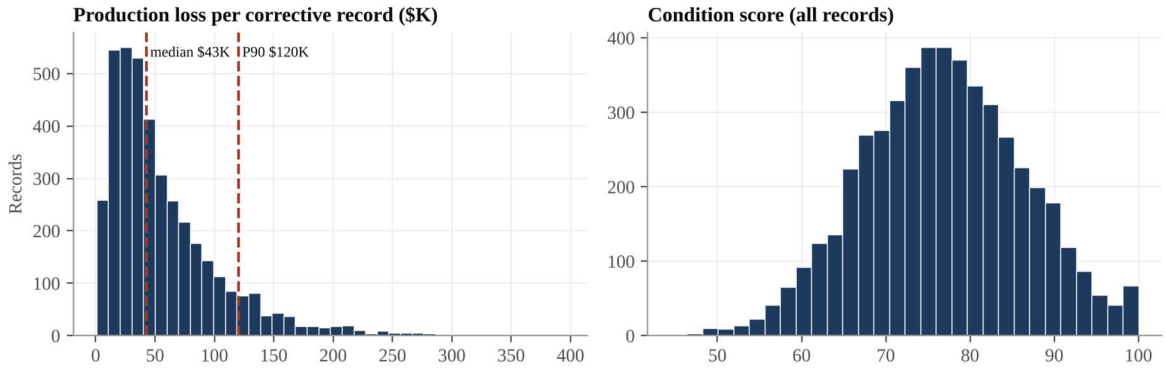


Figure 6.1 Production loss per corrective record (left) and condition score (right).

Production loss is strongly skewed: most corrective jobs lose between $20,000 and $80,000, but a small number lose much more, up to $394,000. The median loss is $42,800, while the most expensive 10% of jobs start at about $120,000 and together make up 29% of all losses. This matters for finance, because planning on the average would underestimate the impact of large events. Condition scores, by contrast, are spread evenly around 77 out of 100.

Using the usual box-plot rule we identified 194 unusually high-loss records and 119 unusually long downtimes. We checked them and found they were consistent (long downtime at a normal cost per hour), so we kept them. These are exactly the events the company most needs to prevent.

6.2 Trends over time

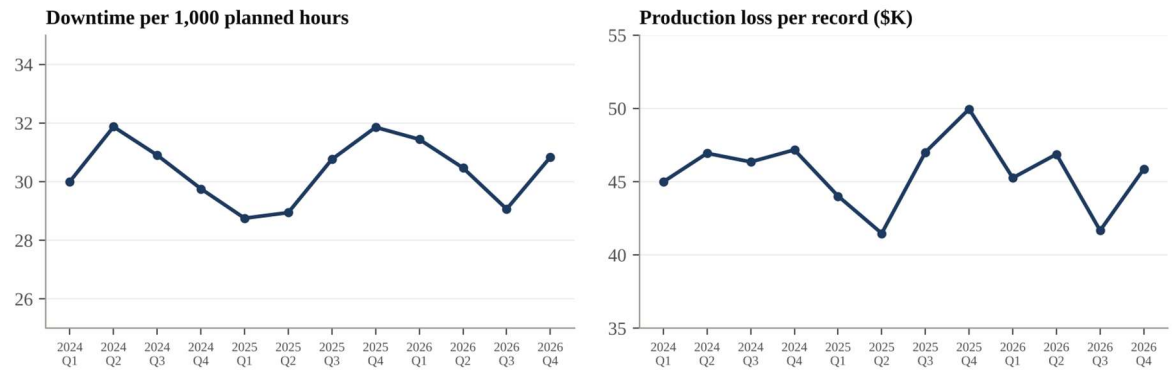


Figure 6.2 Downtime per 1,000 planned hours and production loss per record, by quarter.

We expected to see reliability either improving or getting worse, but the picture is flat. Downtime stays between about 29 and 32 hours per 1,000 planned hours in every quarter, and loss per record stays between $41,000 and $50,000. Loss per record was $46,367 in 2024, $45,611 in 2025 and $44,920 in 2026. Because 2026 has half the records, its totals are lower, but its rates are the same. For management, this means current performance is a stable baseline, and any trial must be compared with a control group rather than with "last year".

6.3 Cost structure: maintenance versus downtime

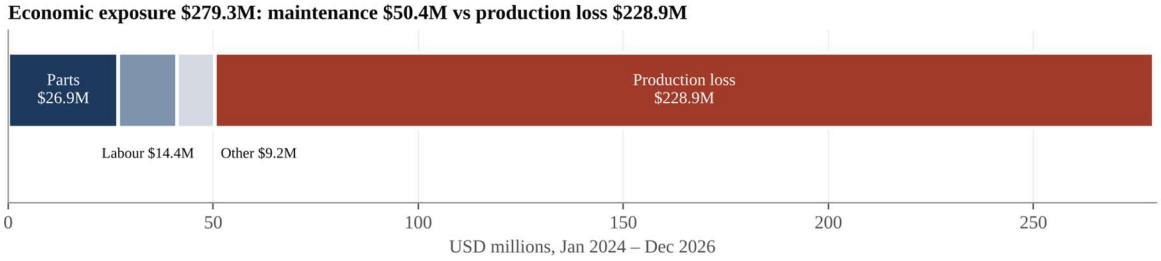


Figure 6.3 Maintenance cost compared with production loss, January 2024 to December 2026.

This was our most important finding. Production loss ($228.9 million) is 4.5 times larger than maintenance spending ($50.4 million) and makes up 82% of the total economic exposure. Each downtime hour costs about $2,513, and this hourly cost hardly changes between asset types ($2,456 to $2,562) or years. In other words, the size of the loss depends almost entirely on how many hours equipment is down. This gives managers a simple way to judge any proposal: how many hours of downtime will it prevent?

Within maintenance spending, parts make up 53%, labor 28% and unexplained "other" costs 18%. Of the maintenance budget, 92% is spent on corrective work after something has already failed. Looking at a single failure makes the imbalance clear: an average failure costs $6,638 to repair but causes $30,128 of lost production.

6.4 Comparing asset classes

Table 6.1 Key results by asset class (rates calculated from totals)

Asset class	MTBF (h)	MTTR (h)	Loss per record	Service target met
Railway Signaling	273	12.7	$55,991	41.1%
Industrial Pump	308	12.4	$48,777	45.6%

Asset class	MTBF (h)	MTTR (h)	Loss per record	Service target met
Heavy Equipment	309	12.2	$48,101	45.6%
Power System	326	11.4	$40,484	47.3%
Biomedical Equipment	362	10.9	$35,714	50.9%

Railway Signaling performs worst on every measure: it breaks down most often, takes longest to repair and loses 57% more per record than Biomedical Equipment. Yet each class receives a similar share of the maintenance budget (between 19% and 21%). Spending the same amount on assets with very different risks suggests the budget is not targeted where it is needed most.

6.5 Criticality: the problem is repair time

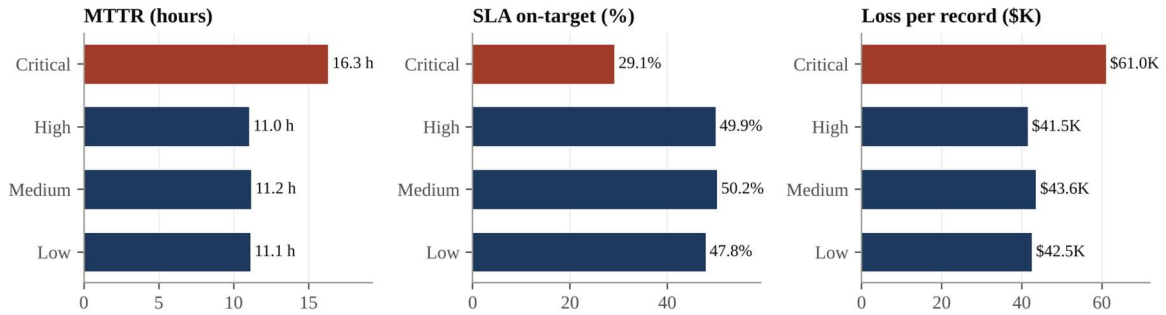


Figure 6.4 Repair time, service performance and loss per record by criticality level.

We expected critical assets to fail more often, but they do not: their MTBF (320 hours) is similar to the other levels. What is different is how long they stay down. Each failure of a critical asset takes 16.3 hours to fix compared with about 11 hours for the others, and only 29% of critical jobs meet their service target. As a result, each critical record loses $61,015 compared with $42,602 for the rest. Critical assets also receive the least spending on spare parts relative to the loss they cause. This points to slow repair, possibly because of parts, escalation or skills, rather than poor prevention.

The size of this problem can be estimated directly. Critical assets have about 508 failures a year, and each one takes about 5.2 hours longer to repair than on other assets. That adds roughly 2,660 hours of downtime a year, worth about $6.7 million of lost production at $2,530 per critical downtime hour. Critical assets receive 9.0 cents of spare parts per dollar of loss, against 12.6 cents for the rest; closing this gap would cost about $0.81 million a year.

6.6 Maintenance strategy

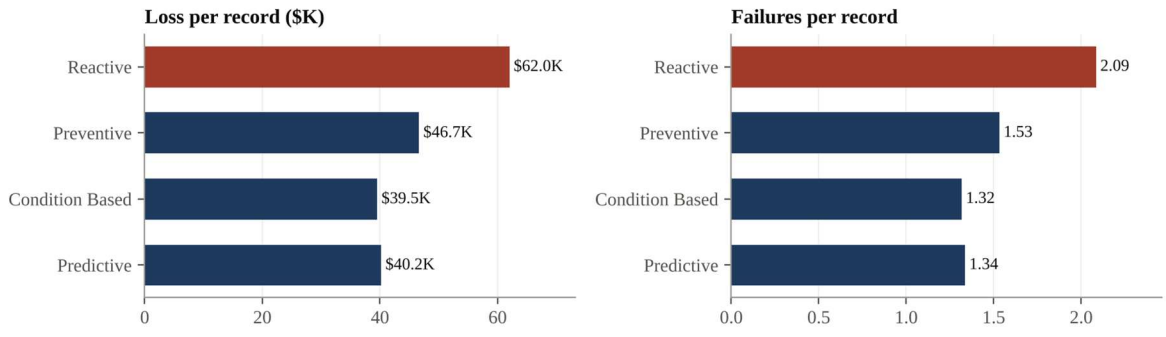


Figure 6.5 Loss per record and failures per record by maintenance strategy.

Reactive maintenance has the worst results: about $62,000 of loss per record and 2.09 failures per record, compared with about $40,000 and 1.3 failures for condition-based and predictive maintenance. However, strategies are not assigned at random. We found that assets under condition-based and predictive care were in better condition to begin with (average score 80 against 73). To compare like with like, we used a regression model that takes age, condition, criticality and asset class into account. The gap became smaller but did not disappear: reactive records still lose about $15,000 to $17,000 more than the other strategies. Full results are in Appendix C. Reactive care is not cheaper to run either: reactive records fail 58% more often than condition-based ones (2.09 against 1.32 failures per record) and cost about $12,300 to maintain against $9,400.



Figure 6.6 Average loss per record by criticality and strategy (thousands of dollars).

Figure 6.6 combines the two factors. The worst combination is critical assets under reactive maintenance, at $89,774 per record, about twice the fleet average of $45,776.

6.7 Asset age and condition

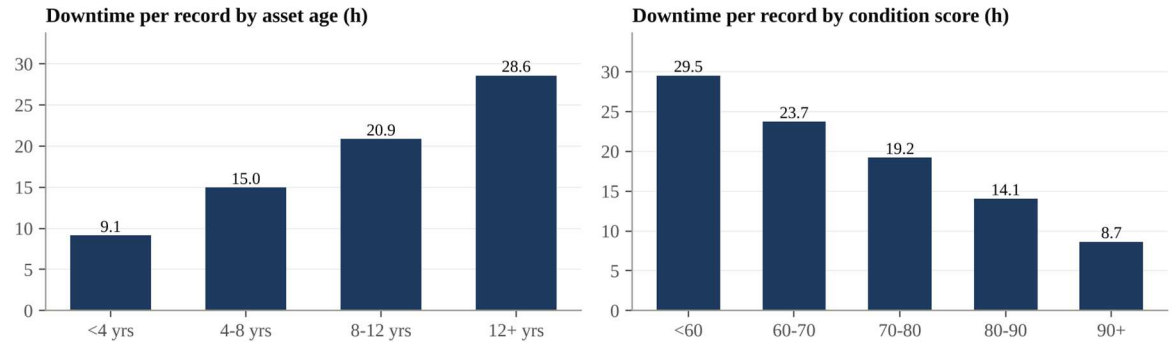


Figure 6.7 Average downtime per record by asset age and by condition score.

Both age and condition show a clear pattern. Downtime per record rises from 9.1 hours for assets under four years old to 28.6 hours for assets over 12 years, and loss rises from $22,800 to $73,200 per record. Maintenance spending, however, increases by only 57% over the same range. Condition score shows the opposite pattern: assets scoring below 60 average 29.5 hours of downtime, while those scoring 90 or more average only 8.7 hours. This suggests that older and worn assets are not receiving maintenance in proportion to the losses they cause.

6.8 Relationships between variables

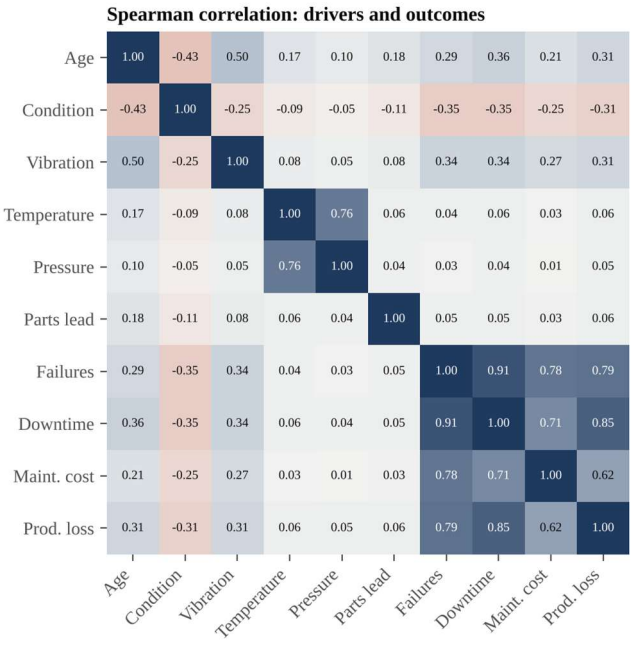


Figure 6.8 Correlations between possible drivers and outcomes (Spearman rank correlation).

We needed to separate real patterns from relationships that exist only because of how the data is calculated. Failures, downtime and loss are very strongly related (0.79 to 0.85), but this is expected: loss is calculated from downtime, and downtime comes from failures. Among the factors known before a job starts, the strongest links with loss are asset age (+0.31), vibration (+0.31) and condition score (–0.31). Temperature, pressure and parts lead time show almost no relationship.

We also checked region, shift, technician team, manufacturer and failure mode. None of them made a meaningful difference: for example, first-time fix rates are between 76% and 77% for every team and shift, and the five failure modes each cause a similar number of failures (1,462 to 1,559). Decisions should therefore focus on criticality, strategy, age and condition rather than on teams or locations.

6.9 What the exploration told us to do next

- Because loss is very skewed, our model predicts whether a job will be *high-loss* (top 20%) rather than the exact amount.
- Age, condition and vibration are useful warning signs, so we used them as model inputs. Results that are only known after the job were excluded.
- Because trends are flat, a simple forecast may be enough for planning (tested in Section 8.4), and any trial needs a control group.
- Because teams and regions make no difference, our recommendations do not target them.

7. Dashboard Design and Implementation

We built the dashboard in Power BI as one file with a single data model, shared filters and a navigation bar. It adapts the five dashboard ideas in the project reference into eight pages that follow the order in which a manager would ask questions: what is happening, why, where, what it costs, and what to do next.



Figure 7.1 The landing page (Summary) of our dashboard.

Table 7.1 Dashboard pages

Page	Question it answers	Main visuals
Summary	What is the overall picture and what should we decide?	KPI cards, trend line, loss by class, criticality × strategy matrix, list of decisions requested
KPI Scorecard	How is each KPI performing?	18 KPI cards grouped into reliability, cost and execution
Reliability	Which assets and conditions drive failures?	MTBF by class, failure modes, age and condition charts
Work Orders	Where is execution weakest?	Service target by criticality, first-time fix, overdue trend, team × shift tables
Cost & Loss	What does downtime cost?	Cost breakdown, loss by criticality, scatter of downtime against loss
Risk	Which jobs need attention first?	Model watchlist, accuracy by tier, condition profiles, exception list
Method	How should the numbers be read?	KPI definitions, data notes, model limits
Asset Class Detail	What is happening inside one asset class?	Drill-through page with KPIs, trend and records ranked by loss

7.1 Visuals and interactivity

The report uses more than the five required visual types: KPI cards, line charts, bar and column charts, a stacked column chart, matrices with color scales, a scatter chart, tables, slicers and navigation buttons. Users can filter every page by year, asset class, criticality and maintenance strategy, and a line at the top of each page shows the current filter. Each KPI card also shows how the current selection compares with the whole fleet. Right-clicking an asset class on any page opens the drill-through page for that class, which ends in a list of its records ranked by production loss.

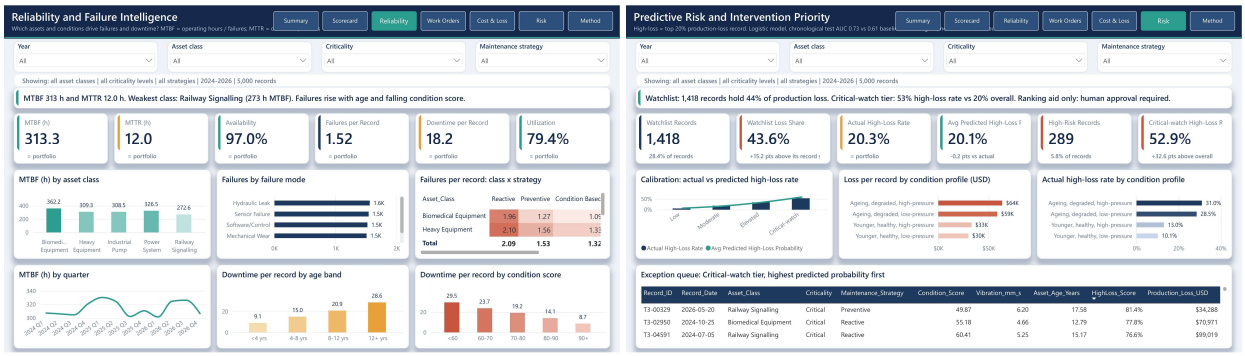


Figure 7.2 The Reliability page (left) and the Risk page (right).

7.2 Measures

All KPIs are DAX measures that divide one total by another, for example:

MTBF (h) = DIVIDE(SUM(Fact[Operating_Hours]), SUM(Fact[Failure_Count]))
Availability = 1 - DIVIDE(SUM(Fact[Downtime_Hours]), SUM(Fact[Planned_Operating_Hours]))
Loss to Maintenance Ratio = DIVIDE([Production Loss], [Maintenance Cost])

Using DIVIDE means that a filter with no failures shows a blank instead of an error. The full list of definitions is in Appendix B.

7.3 How a manager would use it

A manager opening the dashboard first sees that production loss is 4.5 times maintenance spending. Filtering on Critical assets shows the repair time rising to 16.3 hours. The matrix on the Summary page shows that critical assets under reactive maintenance are the most expensive group. Drilling through to Railway Signaling shows its most costly records, and the Risk page gives planners a ranked list of jobs to review.

7.4 Data source and refresh

The CSV file is imported into the Power BI file, so the dashboard works without access to our computers. To refresh it with a new copy of the data: open Transform data, choose Data source settings, select Change Source, point to the new CSV file, then Close & Apply and Refresh. After refreshing, the Summary page should show 5,000 records and $229M of production loss. The model scores are stored as fixed columns, so they only change when the model is re-run and re-validated.

8. AI Applications

The course stresses that the right model depends on the business question, not on which algorithm sounds most advanced. We therefore chose four methods, each linked to one of our business questions, and compared each one with a simple baseline. To make the test realistic, we trained the models on the first 20 months of data (January 2024 to August 2025, 3,340 records) and tested them on the last 16 months (1,660 records) that the models had never seen.

Table 8.1 Overview of AI applications

Method	Business question	Status	Result compared with baseline
Classification: high-loss risk	Which jobs should be reviewed first? (Q7)	Carried out	Clearly better; recommended as a planning aid

Method	Business question	Status	Result compared with baseline
Regression: downtime hours	How long will a job's downtime be? (Q5)	Carried out	Slightly better; not precise enough to use
Clustering: condition profiles	Which groups of assets need different care? (Q8)	Carried out	Useful for description only
Forecasting: quarterly downtime	How much downtime should we plan for? (Q2)	Carried out	A simple average works best
Optimisation of resources	How should technicians and parts be allocated?	Proposed only	Not carried out

8.1 Classification: predicting high-loss jobs

Target. We labelled a record as "high-loss" if its production loss was in the top 20% (above $75,594). This threshold was set using the training period only.

Inputs. We only used information a planner would have before the work starts: asset class, criticality, maintenance strategy, age, planned hours, condition score, vibration, temperature, pressure, energy use, parts lead time and service target. We deliberately left out downtime, cost and the supplied Asset Health Index. The Asset Health Index would have made the model look much better, but it is partly calculated from results such as failures, so using it would have been cheating.

Method and baseline. We used logistic regression, a simple and explainable model. As a baseline we used the historical high-loss rate for each combination of asset class and criticality, which is the kind of rule a planner might use today. We also tried a more complex model (gradient boosting) for comparison.

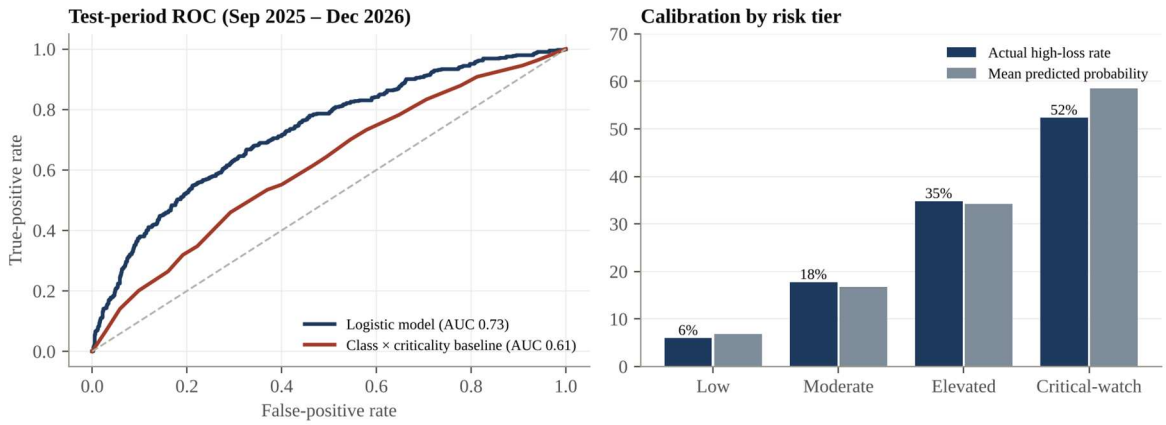


Figure 8.1 Left: how well each approach separates high-loss from normal jobs on the test period (ROC curve). Right: share of jobs that actually became high-loss, by model risk tier.

Results. We measured performance with the AUC score, which shows how well a model ranks risky jobs above safe ones (0.5 is a coin flip, 1.0 is perfect). Our model scored 0.73 on the test period, against 0.61 for the baseline, and the difference is statistically reliable. The more complex model scored 0.71, so it did not justify its extra complexity. The model's risk tiers also behave sensibly: in the highest tier ("Critical-watch") 53% of jobs became high-loss events, compared with 20% overall. The watchlist (the two highest tiers) contains 1,418 records, 28% of the total, but 44% of all production loss, about $99.8 million of the $228.9 million. The exception queue on the dashboard's Risk page shows the kind of job the model flags first: critical assets, 13 to 18 years old, with condition scores below 61 and high vibration.

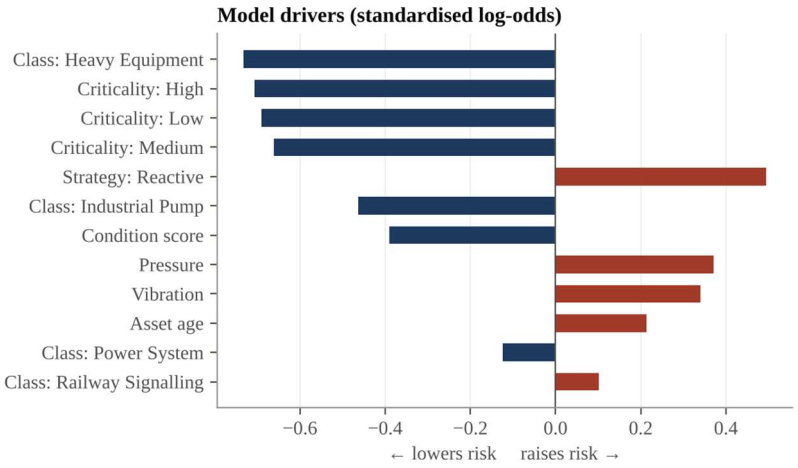


Figure 8.2 Factors that raise (red) or lower (blue) the predicted risk.

Explaining the model. The factors that raise the risk are being a critical asset, reactive maintenance, a low condition score, high vibration and pressure, and older age. These are the same factors we found in the exploratory analysis, which makes the model easy to explain to planners.

Limitations. About half of the jobs in the highest tier do not turn out to be high-loss, so the model will cause some unnecessary inspections. The Maintenance Priority Score already supplied in the data ranks jobs almost as well (also 0.73), so the main advantage of our model is that its inputs are transparent, not that it is more accurate. Finally, predicting risk does not prove that acting on it will save money; that requires a trial. We therefore recommend using the model to support planners, with a person approving every change in priority.

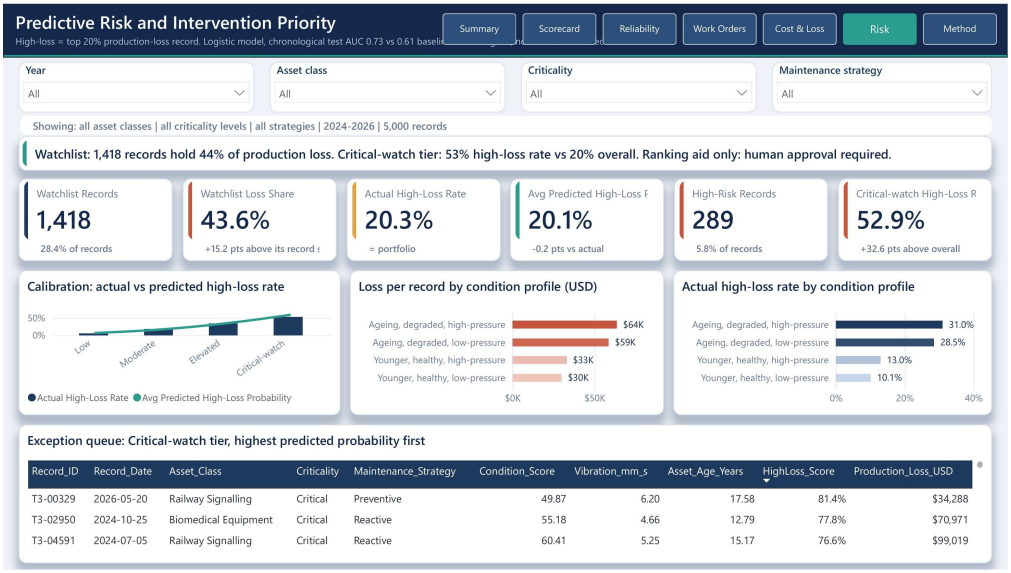


Figure 8.3 The Risk page of the dashboard, showing the watchlist and the list of highest-risk jobs.

8.2 Regression: predicting downtime hours

We also tried to predict the number of downtime hours for each job using the same inputs. Our regression model was wrong by 10.2 hours on average, compared with 11.8 hours for the baseline (the average for each asset class and criticality). The improvement is real but small, given that downtime varies by about 15 hours from job to job. The model is therefore not precise enough to schedule individual jobs, and we did not add it to the dashboard.

8.3 Clustering: asset condition profiles

Following the segmentation approach from the course, we used k-means clustering to group records by asset age, condition score and pressure. We chose four groups because they were easy to interpret, and we checked that the same groups appeared when we re-ran the method several times.

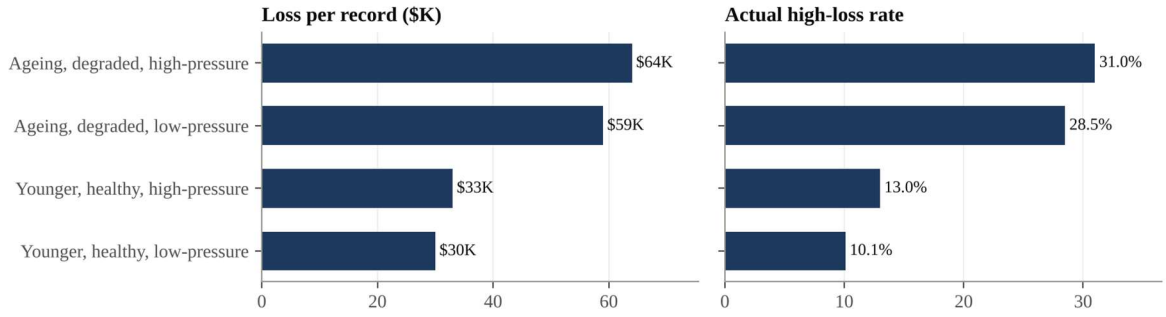


Figure 8.4 Loss per record and high-loss rate for the four condition profiles (values from the dashboard).

The "ageing, degraded" profiles lose about twice as much per record as the "younger, healthy" ones and are about three times more likely to have a high-loss event. The split between high and low pressure mostly reflects asset type (heavy equipment and pumps run at high pressure), so it is not a separate finding. The groups were stable but not very distinct, so we use them to describe the fleet and to choose where to run a trial, not to make predictions.

8.4 Forecasting: how much downtime to plan for

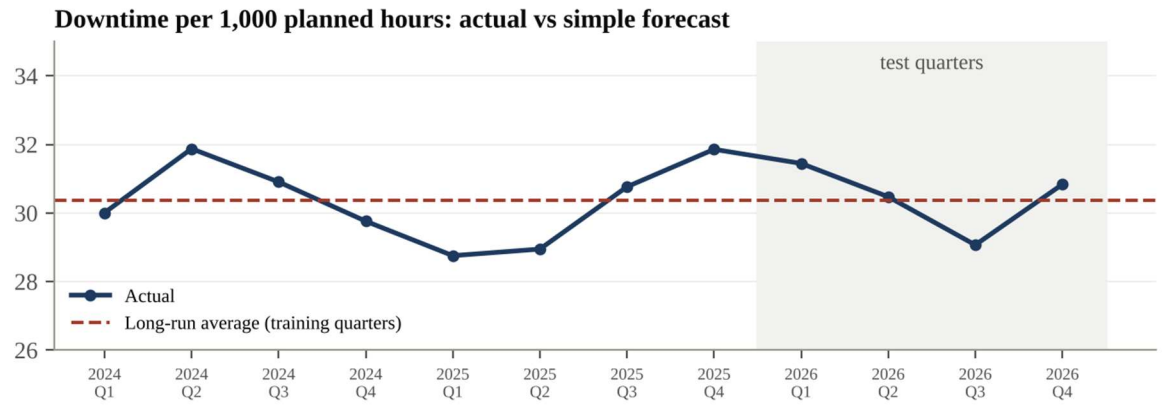


Figure 8.5 Quarterly downtime compared with a simple long-run average forecast.

We compared three simple ways of forecasting next quarter's downtime rate over the last four quarters: using last quarter's value, a straight-line trend, and the long-run average. The long-run average was the most accurate (an average error of 2.5%, against 3.1% for the trend and 3.8% for last quarter's value). This confirms that downtime is stable, and that planning technician capacity on the average is sufficient for now. A more advanced forecasting model would not add value with only 12 quarters of data.

8.5 Proposed but not carried out

An optimization model could decide how to allocate technicians and spare parts to minimise expected loss. This would require a clear objective, real constraints such as technician hours and budget, and a comparison with the current plan. We did not build it, so we make no claim about an "optimal" allocation. Forecasting at the level of individual assets will also only be possible once the asset codes are fixed.

8.6 Model governance

To keep the models trustworthy we used a time-based split, fixed random seeds, set the high-loss threshold on training data only, reviewed every input for leakage, and checked that predicted and actual rates agree. We also propose monthly monitoring of model accuracy and a rule that a person must approve any change in work priority.

9. Executive Insights

The table below brings together our main numerical findings. Each one can be traced to the data and to a page of the dashboard.

No.	Insight	Evidence	What it means for management
1	Downtime is the main cost, not maintenance	$228.9M loss vs $50.4M spending (4.5 times)	Judge every proposal by the downtime it prevents
2	Every downtime hour costs about the same	$2,513 per hour, $2,456–$2,562 by class	One simple value for business cases
3	Performance is stable over time	29–32 downtime hours per 1,000 planned hours each quarter	Trials need control groups
4	High availability hides frequent failures	97.0% availability but MTBF of only 313 hours	Do not rely on availability alone
5	Railway Signaling is the weakest class	MTBF 273 h; $55,991 loss per record	First class to review in detail
6	Critical assets are repaired too slowly	MTTR 16.3 h vs about 11 h; 29% meet target	Focus on faster repair
7	Critical assets get the least parts cover	9 cents of parts per $1 of loss vs 12.6 cents	Protect spare parts for critical assets
8	Reactive maintenance costs most, even after adjustment	About $15,000–$17,000 more loss per record	Test a move to condition-based care
9	Critical assets under reactive care are the worst group	$89,774 per record, twice the average	Start the trial here
10	Older assets lose three times more	$22,800 vs $73,200 per record	Shift maintenance towards older assets
11	Fewer than half of jobs meet the service target	46.1% (not the 75.8% score)	Use "target met" as the service KPI
12	Teams, shifts and regions perform alike	First-time fix 76–77% everywhere	No need for team-specific programs
13	A simple model ranks risk better than a rule	Score 0.73 vs 0.61	Trial it as a planning aid
14	The watchlist concentrates the loss	28% of records hold 44% of loss	Focus inspections there
15	A few large events drive the losses	Top 10% of jobs = 29% of loss	Set reserves for large events
16	The recommendations could pay back quickly	Expected: 2,766 h and $6.56M net a year; 3-year NPV $16.31M	Fund controlled trials to confirm

10. Strategic Recommendations

Following the guidance for executive recommendations in the capstone description, each recommendation below addresses a problem we identified, cites the analytical evidence, recognizes the operating constraints and names a practical owner and next step. We keep three things separate: what the data has **demonstrated**, what we **propose** to do, and any **impact estimate**, which remains unvalidated until it is tested. Recommendations 1 to 4 match the "Decisions requested" panel on our dashboard; 5 and 6 are financial.

Recommendation 1. Speed up repairs on critical assets

Problem addressed	**Critical assets are repaired too slowly (Section 6.5).**
Demonstrated finding	MTTR of 16.3 hours against about 11 hours for other assets; only 29.1% of critical jobs meet the service target; parts spending of 9 cents per $1 of loss against 12.6 cents elsewhere.
Proposed action	Give critical work priority in the repair queue, protect a stock of spare parts for critical assets and agree a clear escalation path.
Constraints	Technician hours are limited, so other work may be delayed; extra spare-parts stock ties up cash.
Owner and next step	Maintenance planners with Procurement. Review all critical work orders within 90 days.
KPI to track	Critical MTTR and share of critical jobs meeting the service target.
Impact estimate	Expected scenario: about 1,330 downtime hours and $2.6 million net a year (Section 10.2); unvalidated.

Recommendation 2. Trial condition-based maintenance on older, worn assets

Problem addressed	**Reactive maintenance and older, worn assets carry the highest losses (Sections 6.6 and 6.7).**
Demonstrated finding	Reactive records lose about $15,000–$17,000 more per record even after adjusting for age, condition, criticality and class; assets over 12 years lose three times more than assets under 4 years; critical assets under reactive care lose $89,774 per record.
Proposed action	Run a six-month trial of condition-based maintenance, starting with critical assets currently under reactive care, and compare them with a similar control group.
Constraints	The pattern comes from synthetic data and is an association, not proof of cause; monitoring equipment and training have a cost.
Owner and next step	Reliability engineers. Select trial and control groups in month 1.
KPI to track	Loss and failures per record, trial group against control group.
Impact estimate	Expected scenario: about 790 downtime hours and $2.5 million net a year (Section 10.2); unvalidated until the trial.

Recommendation 3. Trial the risk watchlist as a planning aid

Problem addressed	**Planners have no forward-looking ranking of risky jobs (Section 8.1).**

Demonstrated finding	The model scores 0.73 against 0.61 for a simple rule; the watchlist holds 28% of records but 44% of production loss.
Proposed action	Run the watchlist next to current planning for three months, with a person approving every change in priority, then decide whether to continue.
Constraints	About half of the highest-risk alerts are false alarms; the supplied priority score ranks almost as well; predicting risk does not prove savings.
Owner and next step	Reliability lead with the Maintenance Director. Monthly check of model accuracy.
KPI to track	Share of loss captured by the watchlist; agreement between predicted and actual rates.
Impact estimate	Expected scenario: about 640 downtime hours and $1.5 million net a year (Section 10.2); unvalidated.

Recommendation 4. Fix data definitions before any roll-out

Problem addressed	**Several data problems limit what can be analysed (Section 5.3).**
Demonstrated finding	Asset codes do not identify real assets; 18% of maintenance cost is unexplained; two different service measures; the stored MTTR field is inconsistent.
Proposed action	Introduce stable asset codes, itemize all maintenance costs, adopt one service measure (target met) and correct the MTTR field.
Constraints	Requires changes in the maintenance and finance systems and agreement between departments.
Owner and next step	KPI lead with the Finance controller. Agree definitions within 60 days.
KPI to track	Number of data-quality checks passed.
Impact estimate	Not a savings measure; it enables all other actions.

Recommendation 5. Base the maintenance budget on downtime

Problem addressed	**The budget is spread evenly while losses are not (Sections 6.3 and 6.4).**
Demonstrated finding	Production loss is 4.5 times maintenance spending; each downtime hour costs about $2,513; each asset class receives 19–21% of the budget although losses differ by up to 57%.
Proposed action	Value every maintenance proposal by the downtime hours it is expected to prevent, multiplied by $2,513.
Constraints	The hourly cost is derived from simulated data and should be checked against real production figures.
Owner and next step	Finance with the Maintenance Director. Apply in the next budget cycle.
KPI to track	Loss to maintenance ratio.
Impact estimate	See Section 10.2: each 1% less downtime is worth about $0.9 million a year.

Recommendation 6. Set reserves for large events

Problem addressed	A small number of events cause a large share of losses (Section 6.1).
Demonstrated finding	The top 10% of jobs cause 29% of loss; the 90th-percentile loss is $120,322 against an average of $57,006 per corrective job.
Proposed action	Use the 90th-percentile event loss, not the average, when setting financial reserves and service commitments.
Constraints	Large-event estimates can change as new data arrives.
Owner and next step	Finance controller. At the next reserve review.
KPI to track	Size of actual large events compared with the reserve.
Impact estimate	Not a savings measure; it reduces financial surprise.

10.1 Priorities

Recommendations 1, 4 and 6 need little investment and can start immediately. Recommendations 2 and 3 should begin as controlled trials, with further budget released only if downtime falls compared with the control group. Recommendation 5 can be applied in the next budget cycle.

10.2 Simulation of financial benefits

To show what the recommendations could be worth, we simulated their effect on downtime, production loss and cost. The simulation is built from the dataset itself: each benefit is calculated from the rates and volumes we measured in Sections 6 and 8, and every cost comes from the data. We agreed the improvement levels as a team and kept them deliberately moderate. The results are estimates for planning and discussion; they are not savings that have been achieved.

Baseline and method

We use the 2024–2025 annual run-rate as the baseline, because 2026 contains only half a year of records. In a typical year the fleet produces about 2,002 records, 36,530 hours of downtime and $92.07M of production loss, against $20.17M of maintenance spending. Benefits are converted between hours and dollars using the cost of a downtime hour measured in the data. Financial results are calculated over three years with a 10% discount rate. For each lever we report downtime hours avoided, gross benefit, cost, net annual benefit, three-year net present value (NPV), return on investment (ROI) and payback period.

Table 10.1 Simulation inputs and where they come from

Input	Value	Source
Annual downtime / production loss (baseline)	36,530 h / $92.07M	Dataset, 2024–2025 average
Cost of one downtime hour (fleet / critical assets)	$2,513 / $2,530	Dataset: loss ÷ downtime hours
Critical failures per year; current and target MTTR	508; 16.4 h → 11.1 h	Dataset; target = MTTR of other criticality levels
Spares uplift needed to reach other tiers' parts cover (12.6¢ per $1 of loss)	$0.81M per year	Dataset: parts spending ÷ loss
Reactive records per year	319	Dataset
Loss gap, reactive vs condition-based (adjusted)	$16,637 per record	Our regression, Appendix C.1
Maintenance cost gap, reactive vs condition-based	$2,995 per record	Dataset: average maintenance cost

Input	Value	Source
Watchlist records per year (not critical, not reactive)	252, of which 84 high-loss	Our model; critical and reactive records excluded to avoid double counting
Loss avoided when a high-loss event becomes a normal one	$96,069	Dataset: average high-loss minus average normal loss
Cost of one inspection	$342	Dataset: 6.5 technician hours per work order × $52.69 per hour

Table 10.2 Improvement levels in the three scenarios

Lever	Conservative	Expected	Optimistic
Share of the critical MTTR gap closed	25%	50%	75%
Spares uplift (share of gap to 12.6¢)	50%	100%	100%
Share of reactive records converted to condition-based	25%	50%	75%
Share of the adjusted loss gap achieved after conversion	50%	75%	100%
Share of flagged high-loss events prevented by inspection	10%	20%	30%

Results

Table 10.3 Annual benefits, costs and three-year financial results by lever and scenario

Lever	Scenario	Hours avoided / yr	Gross benefit / yr	Cost / yr	Net / yr	3-yr NPV	ROI	Payback (months)
Faster critical repairs with spares uplift	Conservative	666	$1.69M	$0.40M	$1.28M	$3.19M	318%	2.9
	Expected	1,332	$3.37M	$0.81M	$2.56M	$6.38M	318%	2.9
	Optimistic	1,998	$5.06M	$0.81M	$4.25M	$10.57M	527%	1.9
Reactive to condition-based maintenance	Conservative	264	$0.90M	$0.00M	$0.90M	$2.24M	n/a*	immediate*
	Expected	792	$2.47M	$0.00M	$2.47M	$6.14M	n/a*	immediate*
	Optimistic	1,584	$4.70M	$0.00M	$4.70M	$11.68M	n/a*	immediate*
Watchlist inspections	Conservative	321	$0.81M	$0.09M	$0.72M	$1.79M	835%	1.3
	Expected	642	$1.61M	$0.09M	$1.53M	$3.80M	1,770%	0.6
	Optimistic	963	$2.42M	$0.09M	$2.33M	$5.81M	2,705%	0.4
Total	Conservative	**1,251**	**$3.39M**	**$0.49M**	**$2.91M**	**$7.22M**	594%	1.7
	Expected	**2,766**	**$7.45M**	**$0.89M**	**$6.56M**	**$16.31M**	735%	1.4

Lever	Scenario	Hours avoided / yr	Gross benefit / yr	Cost / yr	Net / yr	3-yr NPV	ROI	Payback (months)
	Optimistic	**4,545**	**$12.17M**	**$0.89M**	**$11.28M**	**$28.06M**	1,264%	0.9

Gross benefit includes avoided production loss and, for condition-based maintenance, lower maintenance cost. *The dataset shows no extra cost for condition-based maintenance (it is $2,995 per record cheaper), and as agreed we did not add an external sensor cost, so ROI and payback are not meaningful for this lever; see the break-even test below.

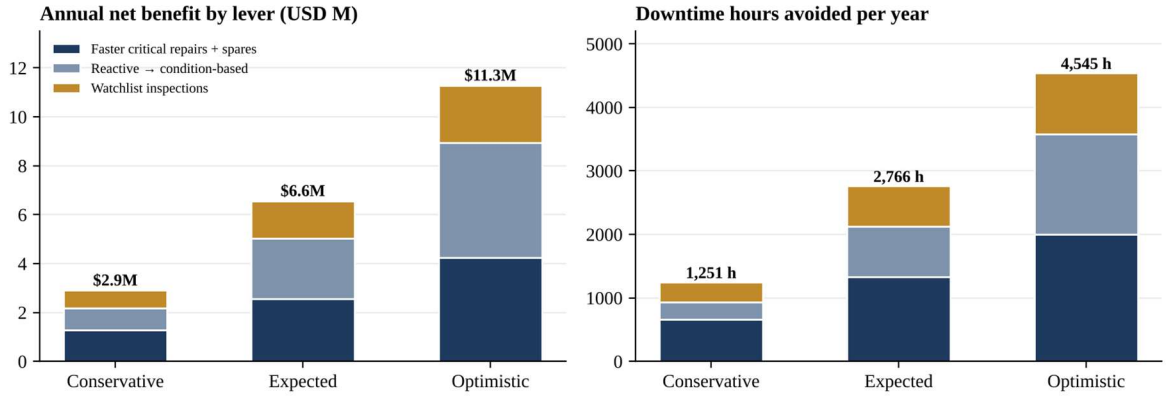


Figure 10.1 Net annual benefit and downtime hours avoided by lever in each scenario.

In the **expected scenario**, the three levers together avoid about **2,766 hours of downtime a year**, about 7.6% of current downtime. This is worth $6.97M of avoided production loss plus $0.48M of lower maintenance cost, against $0.89M of extra spending on spare parts and inspections. The **net benefit is about $6.56M a year**, and the **three-year NPV is about $16.31M**. Even in the conservative scenario the net benefit is about $2.91M a year (NPV $7.22M), and in the optimistic scenario it rises to about $11.28M a year (NPV $28.06M).

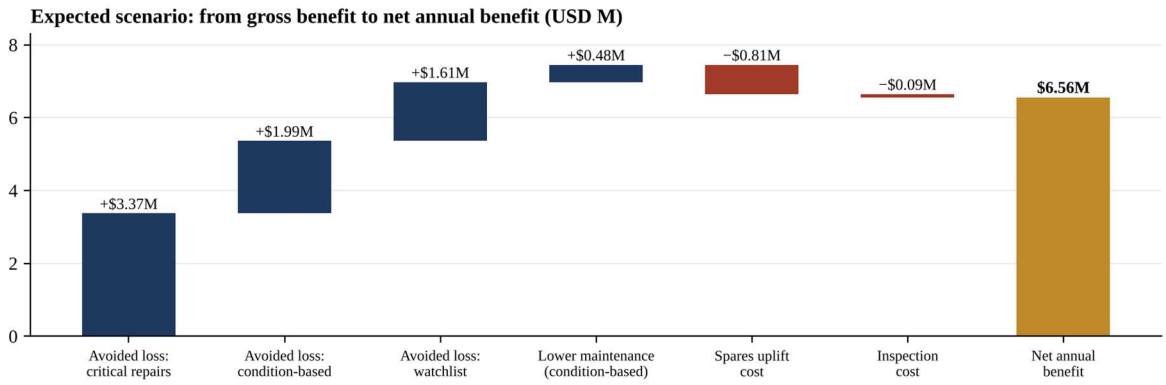


Figure 10.2 Expected scenario: how the gross benefits and costs add up to the net annual benefit.

Faster repair of critical assets is the largest single lever ($2.56M net a year in the expected case), because each critical downtime hour is expensive and the gap to other assets is large. Converting reactive maintenance is the second lever. The watchlist adds the smallest benefit, partly because we excluded critical and reactive records from it to avoid counting the same saving twice, but it is also the cheapest lever, costing about $0.09M a year in inspections.

Break-even tests

- **Spares uplift.** The full uplift ($0.81M a year) pays for itself if it prevents 319 hours of critical downtime, which means closing only 12% of the gap between critical and other assets' repair times.
- **Watchlist.** The inspections ($0.09M a year) pay for themselves if they prevent about one high-loss event a year, or about 1% of the flagged high-loss events.
- **Condition-based maintenance.** Because the data shows no extra cost, we tested how much a sensor and monitoring system could cost before the lever stops paying. In the expected scenario it could cost up to about $15,500 per converted record each year (about $11,300 in the conservative case) and still break even.

Limits of the simulation

The simulation applies rates from synthetic data and assumes the agreed improvement levels can be reached; it does not prove that they will be. Benefits start in full in year one, so real results would be lower while changes are being introduced. Condition-monitoring equipment, training and programme management are not included, as agreed, so real net benefits will be lower than shown. The three levers were designed not to overlap, but some interaction is still possible. For these reasons, Section 10.1 recommends releasing budget only after a controlled trial confirms the downtime reduction.

10.3 Linking to implementation

The course highlights that analytics projects often fail because of data quality, missing skills and resistance to change rather than because of the models. Our plan reflects this: we start with data definitions (Recommendation 4), keep people in control of decisions, and prove results with small trials before asking staff to change how they work. Over time, these responsibilities could sit within a small analytics competency team that owns the KPI definitions, the dashboard and the model monitoring. To show how this could work in practice, the team also built a prototype platform, Asset Reliability, which repeats the analysis with AI agents, the same KPI definitions and a human approval step before anything is published. It is described in a separate appendix; it uses sample data and none of the figures in this report come from it.

10.4 What we do not recommend

We do not present any of these recommendations as an optimal decision. Our rankings are descriptive, and an optimization claim would need an explicit objective, realistic constraints such as technician hours and budget, and a solved comparison with the current plan, which we have not done (Section 8.5). We also do not recommend changing the maintenance strategy for the whole fleet based on this analysis alone, letting the model change priorities automatically, acting on individual assets or sites before the asset codes are fixed, or creating programs for specific teams or regions.

11. Conclusion

This project showed us that the biggest cost in this company's asset base is not what it spends on maintenance but what it loses when equipment stops. Production loss is 4.5 times maintenance spending, and every hour of downtime costs about $2,513. The loss is concentrated in identifiable places: critical assets that take too long to repair, reactive maintenance, and older or worn equipment, with Railway Signaling the weakest class.

Our dashboard allows managers to see these patterns and drill down to the records behind them. Our predictive model ranks risky jobs better than a simple rule, but it is not more accurate than the score already in the data and it does not prove that acting on it will save money. We think this honest result is still valuable: it shows that the model can support planners, as long as people stay responsible for the final decision.

Our simulation suggests that the recommended actions could avoid around 2,766 hours of downtime a year, worth a net $6.56M a year in the expected scenario, but this still needs to be proven in practice.

Decision implications. Management can act now on critical-asset repairs, data definitions and reserves, and should test condition-based maintenance and the watchlist through controlled trials measured in hours of downtime avoided.

Further evidence needed. Before a real roll-out, the company would need stable asset codes, production loss measured independently of downtime, timestamps showing when condition readings are taken, and results from a trial with a control group.

In line with the course theme of moving from descriptive to prescriptive analytics, we believe predictive maintenance creates value only when reliable warnings lead to timely, funded and approved action on the right asset before it fails.

12. AI Usage Disclosure and Declaration

This project involves two different kinds of AI, and we keep them separate. The **analytical models** (logistic regression, ridge regression and k-means clustering) are part of our analysis and are evaluated in Section 8. **Generative AI assistants** were used to support the team's work, as described below.

12.1 Tools used and scope of assistance

Tool	Purpose	Scope of assistance
Claude (Anthropic), used through Claude Cowork	Analysis support and drafting	Helped write and run Python code for the data checks, KPI calculations, exploratory charts, model testing and the financial simulation; helped draft, format the report.
AI coding assistant	Software development	Helped write code for the Asset Reliability prototype platform, which is described in a separate appendix and is not the source of any figure in this report
Language models inside the prototype (Claude, OpenAI, Ollama)	Optional platform features	Designed to phrase verified findings and raise questions for a human approver; not active in the production deployment and not used to produce any result in this report
Microsoft Power BI Desktop	Dashboard	Built by the team using standard Power BI features
MCP Claude Code	Analysis and charts	Open-source tools used to run the analysis

12.2 How we validated the work

- All totals and KPIs were recalculated from the original CSV and compared with the workbook and the Power BI dashboard (Appendix D).
- The model was re-run and its results matched those shown on the dashboard; every model input was checked to make sure no information from after the job was used.
- Every number in the report, the presentation and the speaker notes was checked against our tables, and every savings figure is labelled as an estimate.

- No AI-generated text, invented figure or unverified model result is presented as analytical evidence. The synthetic nature of the data and the limits of generalization are stated throughout.

12.3 Declaration

1. The AI tools used in this project and the purpose of each use are fully disclosed above.
2. All analytical results in this report were produced from the assigned dataset and checked by the team; none were generated by a language model.
3. We reviewed and understood all content prepared with AI support, and each of us can explain the decisions, evidence and limitations, including the parts led by other members.
4. Final analytical judgement, the conclusions and the recommendations are our own, and we take full responsibility for this submission.

References

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Appendix A. Team Contribution

Team member	Reg. No.	Main contribution	Presentation slides
Malak Philippe Sadek Reiad	24220177	Project management, business questions, data governance and integration of the report and presentation	1–3
Ahmed Fawzy Mohamed Rashed	24220998	Power BI data model, measures and dashboard pages; dashboard demonstration	4, 5, 15
Ahmed Abd Elhakam Abd Elfattaah Zaian	24224008	Automation and AI use: AI method design, leakage review and the Asset Reliability prototype platform	8, 10, 14

Team member	Reg. No.	Main contribution	Presentation slides
Khaled Abdelnasser Morsi Amer	24221044	Risk model and baseline comparison; financial benefit simulation	9, 11, 12
Ali Elraey	24126750	Reliability analysis (MTBF, MTTR), spare parts and procurement review; recommendations	6, 7, 13

All members reviewed the final dashboard, report and presentation and can explain the full analysis, including the parts led by others.

Appendix B. KPI Definitions and Glossary

KPI	Formula	Fleet value
Availability	1 − total downtime hours ÷ total planned hours	97.0%
Utilization	Total operating hours ÷ total planned hours	79.4%
MTBF	Total operating hours ÷ total failures	313.3 h
MTTR	Total downtime hours ÷ total failures	12.0 h
Maintenance cost	Sum of maintenance cost	$50,430,446
Production loss	Sum of production loss	$228,881,052
Loss to maintenance ratio	Production loss ÷ maintenance cost	4.54
Downtime cost rate	Production loss ÷ downtime hours	$2,513
Cost per operating hour	Maintenance cost ÷ operating hours	$21.19
Loss per record	Production loss ÷ number of records	$45,776
Overdue rate	Overdue work orders ÷ all work orders	14.1%
First-time fix rate	First-time fixes ÷ corrective work orders	76.6%
Service target met	Records with actual hours ≤ target hours ÷ records	46.1%
Watchlist	Records in the Elevated or Critical-watch risk tiers	1,418 (28.4%)

Glossary. *AUC*: a score from 0.5 (random) to 1.0 (perfect) showing how well a model ranks risky cases above safe ones. *Baseline*: a simple rule used to judge whether a model adds value. *Data leakage*: giving a model information it would not have at the time of the decision. *k-means clustering*: a method that groups similar records together. *Logistic regression*: a model that estimates the probability of an event. *Percentile (P90)*: the value below which 90% of cases fall. *Reactive maintenance*: repairing equipment only after it fails. *Condition-based maintenance*: acting when measurements show signs of wear.

Appendix C. Validation and Model Evidence

C.1 Strategy comparison adjusted for other factors (regression on loss per record)

Strategy compared with reactive	Difference in loss per record	95% confidence interval
Condition Based	−$16,637	−$20,662 to −$12,611
Predictive	−$15,889	−$20,063 to −$11,714
Preventive	−$14,989	−$18,686 to −$11,292

Controls: asset age, condition score, criticality and asset class ($R^2 = 0.16$).

C.2 Classification model (test period, 1,660 records)

Approach	AUC	95% interval	Top-10% lift
No model	0.50	–	1.0×
Baseline: rate by class and criticality	0.61	0.58–0.65	1.8×
Logistic regression (chosen)	0.73	0.70–0.76	2.5×
Gradient boosting	0.71	–	–
Supplied Maintenance Priority Score	0.73	–	–

The improvement over the baseline is between +0.08 and +0.15 (95% bootstrap interval, 2,000 samples). Risk tiers: Low below 10%, Moderate 10–25%, Elevated 25–50%, Critical-watch above 50%. Actual high-loss rates by tier: 5.7%, 18.1%, 34.0% and 52.9%. Excluded inputs: Asset Health Index (AUC 0.82, but includes outcome information) and Failure Probability (fully determined by other inputs).

C.3 Other models

Model	Result	Baseline
Downtime regression (ridge)	Average error 10.2 h (R^2 0.28)	11.8 h (class and criticality average)
k-means clustering, 4 groups	Silhouette 0.30; stability (ARI) 0.96 across repeated runs	–
Quarterly forecast	Long-run average: 2.5% average error	Last value 3.8%; linear trend 3.1%

Appendix D. Reconciliation with the Dashboard

Figure	Report	Dashboard	Match
Availability / MTBF / MTTR	97.0% / 313.3 h / 12.0 h	97.0% / 313.3 / 12.0	Yes
Maintenance cost / production loss	$50.4M / $228.9M	$50M / $229M	Yes (rounded)
Loss to maintenance / downtime cost rate	4.5 / $2,513	4.5x / $2,513	Yes
Overdue / first-time fix / service target met	14.1% / 76.6% / 46.1%	14.1% / 76.6% / 46.1%	Yes
Critical MTTR / service target met	16.3 h / 29.1%	16.3 h / 29.1%	Yes
Loss per record by strategy	$62,030 / $46,658 / $39,543 / $40,216	Same	Yes
Downtime per record by age band	9.1 / 15.0 / 20.9 / 28.6 h	Same	Yes
Model AUC / baseline	0.73 / 0.61	0.73 / 0.61	Yes
Watchlist / share of loss	1,418 / 43.6%	1,418 / 43.6%	Yes

Appendix E. Learning Outcomes and Where They Are Shown

The table below shows where this report and its companion files show each learning outcome from the capstone description.

Learning outcome	Where it is shown
Translate a business scenario into explicit executive questions and decision objectives	Sections 2.1–2.5 (decision, stakeholders, approach) and 3 (eight business questions linked to decisions)
Interpret the dataset, its grain, dimensions, information timing and synthetic assumptions	Section 4 (grain, dimensions, timing of variables) and 2.6 (synthetic data)
Validate completeness, uniqueness, business rules, accounting identities and KPI formulas	Section 5.2–5.3 (checks and problems found); Appendix D (reconciliation)
Conduct systematic EDA of distributions, trends, relationships and anomalies	Section 6.1 (distributions and outliers), 6.2 (trends), 6.3–6.7 (performance by segment), 6.8 (relationships)
Engineer and validate derived measures with correct denominators and financial scope	Section 5.4 (rates from totals), 5.5 (new fields); Appendix B (KPI formulas)
Build interactive Power BI dashboards with a coherent data model, navigation and drill-through	Section 7 and the Group_9226_Team_3_Dashboard.pbix file
Connect descriptive and diagnostic analysis with predictive or prescriptive applications	Section 2.2 (analytics maturity), 6.9 (how EDA shaped the models), 8 and 10
Select and evaluate AI methods against informative baselines with appropriate validation	Section 8 (time-based split, baselines for every model); Appendix C
Explain model performance, uncertainty and practical limits in business language	Sections 8.1–8.4 (results and limitations in plain terms); Appendix B glossary
Develop recommendations that recognize operating constraints and the evidence needed	Section 10 (constraints, owners, trials, what we do not recommend) and 11 (further evidence needed)
Communicate findings through a professional report and a concise executive briefing	This report and Group_9226_Team_3_Presentation.pptx
Use AI tools responsibly and disclose their contribution	Section 12 (AI Usage Disclosure)